

R Project

Use an R Notebook to perform analyses on the **PerioData.csv**. Use the R Cookbook to figure out how to do all the analyses. Use the markup language to indicate what analyses have been done. NAME your notebook, ***YourLastNameRProject.Rmd*** and load the finished notebook (the actual notebook) to Canvas. *Let me repeat that: TURN IN THE “.Rmd” FILE.*

NOTE: You only have to import a dataset and make a dataframe one time in the file. Also, make sure only to read in the file using only the filename “PerioData.csv”, not a path to the file that only exist on your computer. More information on the PerioData.csv can be found on this pdf, page 5: <https://kelleybioinfo.org/algorithms/basics/programming/r.pdf>

Part A: Univariate statistics with R

Normality check & data transformation

- (1) Make histograms, run qqnorm and qqline of all bacteria and “deepest”.
- (2) Transform variables if non-normal and repeat histogram, qqnorm, qqline on transformed data. For transformations I suggest trying either sqrt() or log().
- (3) Example: leptos=sqrt(leptos)

For the following, used the normalized (transformed) values.

One-way ANOVA and summary and boxplots:

- (1) Lepto by time
- (2) Strep by time

Correlation analysis (cor.test)

- (1) Strep vs. prev
- (2) Strep vs. fuso
- (3) Fuso vs. leptos

Linear regressions

Use the lm command and use plot and abline to produce graphs.

- (1) Strep vs. deepest
- (2) Fuso vs. deepest

Part B: Use ggplot2 to make pretty graphs

First, load the ggplot2 library and dependencies. It may require installation.

Helpful tutorial:

<https://www.tutorialspoint.com/ggplot2/index.htm>

Produce the following graphs:

- (1) ggplot histogram of prev variable.
- (2) ggplot scatterplot of strep by deepest, colored by time
- (3) ggplot scatterplot of fuso by deepest
- (4) Same as 3 but use smoothing algorithm

Part C: rclr transform the variables (just the bacteria) and repeat all the statistical tests above.

Part D: Multidimensional scaling before and after rclr transformation

Follow the directions in the **mds_code.Rmd** file which performs multidimensional scaling.

Rename the file **YOURLASTNAMEmds_code.Rmd** which you will turn in.

In the exercise, you will be doing doing beta-diversity analysis with the relative abundance data and then also with the rclr transformed data.